

Theorem 1.1 - Infinite Instability (Finite-Time Blow-up) for Bubble Maximus SDE

SDE Formulation of 3D Spacetime as Maximus Stability Bubble

Nasir Khan | ORCID: 0009-0002-6006-0976 |
nasirk684@gmail.com

Wana, Pakistan | DOI: 10.5281/

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ABSTRACT:

We prove finite-time infinite instability for bubble radius SDE governing Bubble Maximus. SDE $dR_t = (-k_1/R_t^2 + k_2 R_t^3)dt + \sigma R_t dW_t$ exhibits almost-sure finite-time explosion, divergent second moment, and infinite Lyapunov exponent. Physical regularization yields $T_{\text{gas}} \sim 10^4 \text{ K}$ (sonoluminescence) and $P_{\text{jet}} \rightarrow \text{infinity}$

(cavitation damage), linking mathematical infinity to observable light.

1. THEOREM 1.1 - INFINITE INSTABILITY

Consider SDE on filtered probability space $(\Omega, \mathcal{F}, P)$:

$$dR_t = (-k_1 / R_t^2 + k_2 R_t^3) dt + \sigma R_t dW_t, t \geq 0$$

$R_t > 0$, $k_1 > 0$ surface tension, $k_2 > 0$ Maximus super-linear growth,

$\sigma > 0$ noise, (W_t) Brownian motion

Initial: $R_0 = r_0 > 0$ finite deterministic

Definition of explosion time:

$$T^* = \inf\{ t \geq 0 : R_t = 0 \text{ or } R_t = +\infty \}$$

2. RESULT

(i) First-hitting explosion: $P(T^* < \infty) = 1$

Process explodes in finite time almost surely.

(ii) Divergence: $E[R_t^2] = +\infty$ as $t \rightarrow T^*$

Second moment blows up.

(iii) Lyapunov exponent:

$\lambda = \lim_{t \rightarrow T^*} (1/t) \log R_t = +\infty$ a.s.

Exponential instability dominates.

3. PROOF SKETCH

Step 1 - Deterministic collapse for small R :

For $R \leq \epsilon$, $f(R) \leq -k_1/(2 R^2)$

ODE $dR/dt = -k_1/R^2 \rightarrow$ solution $R(t) = (R_0^3 - 3k_1 t)^{1/3}$

Collapse time $T_c = R_0^3 / 3k_1 < \infty$

Step 2 - Super-linear blow-up for large R :

For $R \geq M$, $f(R) \geq (k_2/2) R^3$

ODE $dR/dt = (k_2/2) R^3 \rightarrow$ blows up at $T_b = 1/(k_2 R_0^2) < \infty$

Step 3 - Stochastic bound:

Let $Y(t) = E[R_t^2]$. Using Ito formula and Jensen $E[R^4] \geq (E[R^2])^2$:

$dY/dt \geq C Y^2$ for $C > 0$

Riccati ODE gives $Y(t) \geq 1 / (1/Y_0 - C t)$

Thus $Y(t) \rightarrow +\infty$ at finite time $T^* = 1/(C Y_0)$

Step 4 - Brownian motion ensures visit:

(W_t) a.s. visits $(0, \epsilon)$ union $(M, \infty) \rightarrow$ combines with

deterministic bounds $\rightarrow$ explosion with probability 1.

Thus $P(T^* < \text{infinity})=1$, $E[R^2]$ diverges,
 $\lambda=+\text{infinity}$. QED.

4. PHYSICAL REGULARIZATION - MATH INFINITY = PHYSICAL LIGHT

Energy density $e = E_0 / (4/3 \pi R^3)$ proportional
 $R^{-3} \rightarrow \text{infinity}$ as $R \rightarrow 0$

But real gas heats:

$$T_{\text{gas}} = T_0 (R_{\text{max}} / R_t)^{3(\gamma-1)} \sim 10^4 \text{ K}$$

At $T^* \rightarrow 10^4 - 2 \times 10^4 \text{ K} \rightarrow \text{ionization} \rightarrow \text{plasma}$
 $\rightarrow h \nu$ sonoluminescence flash.

Jet pressure:

$$P_{\text{jet}} \approx \rho v^2 \rightarrow \text{infinity} \text{ as } R_t \rightarrow 0+$$

Explains cavitation damage and material pitting.

Interpretation: Math infinity = Physical light + damage.

Singularity is regularized by observable physics.

5. CONNECTION TO MAIN PAPER

This theorem proves the dynamics claimed in main paper SDE-Physics-Bubble-Maximus v1.0.

Bubble Maximus SDE is not eternally stable - it has finite-time infinite instability.

Our universe lives in Kramers residence time before blow-up.

Flow: Instability (noise) $\rightarrow$ Bubble Maximus (metastable, V_{3D}) $\rightarrow$

Black Hole ($V \rightarrow -\infty$, grave M, Q, J) $\rightarrow$ Hawking noise.

White Hole FORBIDDEN.

Kramers lifetime:

$$\tau = 2\pi / \sqrt{V''(\min) |V''(\max)|} * \exp(\Delta V / D)$$

13.8 Gyr = τ since nucleation.

6. FIGURES FOR GITHUB

All 3 figures are uploaded separately along with this PDF:

Figure 1: Stability Potential Landscape - Infinite Instability -> Bubble Maximus (V_{3D}) -> Black Hole ($V \rightarrow -\infty$)

Figure 2: Eternal Nucleation Timeline - Past -> Bubble A -> Bubble B (Our Universe) -> Bubble C -> Future

Figure 3: Theorem 1.1 - Infinite Instability (THIS THEOREM) - theorem_1_1_fig3.png - upload as

additional fig

7. CONCLUSION

We proved Theorem 1.1. Bubble Maximus undergoes finite-time blow-up almost surely. Physically it is seen as sonolum